EXECUTIVE BRIEFING — CONFIDENTIAL

Aurelius

A full-stack AI assistant platform built from first principles. Pure PyTorch. Production-ready. Enterprise-hardened.

1.395B	1,720+	32K+	0
PARAMETERS	MODULES	TESTS	HIGH CVES

Aurelius Systems, Inc. • 2025 • github.com/S3nna13/Aurelius

01 Executive Summary

What Aurelius is, why it matters, and where it is going.

The Elevator Pitch

Aurelius is a vertically integrated AI platform that goes from a 1.4-billion-parameter transformer — trained entirely from scratch in pure PyTorch — all the way to a production-grade API server and a modern React dashboard. Unlike competitors who stitch together third-party libraries, every algorithm is handwritten. This gives us full control over performance, security, and intellectual property.

157+	33	3	28
DEVELOPMENT CYCLES	SUB-SYSTEMS	LANGUAGE RUNTIMES	CVES CLOSED

Aurelius sits at the intersection of **research** and **product**. On one side, it is a deep research codebase where novel architectures (Mamba, Mixture of Experts, State Space Models, diffusion language heads) can be prototyped and benchmarked without framework lock-in. On the other side, it is a deployable conversational AI with memory, multi-step tool use, safety guardrails, and a polished web interface that non-technical users can operate.

The platform is released under the **Aurelius Open License**, which grants broad usage rights while preserving the underlying architecture as the company's intellectual property. This creates a dual benefit: community adoption drives ecosystem growth, while the core orchestration model, memory hierarchy, and design patterns remain competitively protected.

Full Stack Ownership

From transformer weights to pixel-perfect UI, every layer is built in-house. No black-box dependencies on OpenAI, Anthropic, or HuggingFace at runtime.

Security-First Design

28 CVEs tracked and closed. Bandit static analysis, pip-audit dependency scanning, adversarial robustness testing, and a full STRIDE threat model.

Performance at the Edge

Rust native modules for hot paths (token counting, data engine). Speculative decoding, continuous batching, and paged KV cache for throughput.

Future-Proof Architecture

FamilyManifest + ModelVariant pattern lets us maintain base, chat, coding, long-context, retrieval, agent, and safety variants under one contract.

02 The Product

What Aurelius does and who it serves.

Three Entry Points, One Platform

Terminal Chat

A developer-friendly command-line interface. Install with `pip`, type `aurelius chat`, and start conversing. Built-in personas (coding, security, research) adapt tone and expertise on the fly.

API Server

An OpenAI-compatible HTTP API that any existing client can drop into. WebSocket streaming for real-time token delivery. Session routing, rate limiting, and semantic memory built in.

Web Dashboard

A "Mission Control" React application with real-time metrics, agent management, memory browser, workflow designer, and system health monitoring. Dark-mode native, PWA-ready.

Target Use Cases

- ✓ **Enterprise AI Assistant** — Deploy on-premise with full data sovereignty. No third-party API calls required.
- ✓ **Research Platform** — Swap attention mechanisms, test novel architectures, and benchmark against 100+ metrics without framework migration.
- ✓ **Agent Orchestration** — Multi-step tool use with ReAct reasoning, budget-bounded execution, and automatic error recovery.
- ✓ **Security Operations** — MITRE ATT&CK classification, adversarial suffix detection, and red-team probe suites for AI safety.
- ✓ **Code Intelligence** — Built-in coding persona with AST analysis, patch synthesis, and sandboxed code execution.
- ✓ **Knowledge Management** — Layered semantic memory (L0–L4) with importance scoring, decay, and associative recall.
- ✓ **Workflow Automation** — Visual workflow designer with cron scheduling, dependency graphs, and execution monitoring.
- ✓ **Federated Deployments** — Docker, Kubernetes, and Ollama packaging for edge, cloud, and air-gapped environments.

Why "Pure PyTorch" Matters to the Business

Every major competitor relies on a stack of third-party libraries (HuggingFace Transformers, FlashAttention, DeepSpeed, vLLM, etc.). Each dependency is a supply-chain risk, a licensing constraint, and a performance ceiling. By writing every algorithm from scratch, Aurelius eliminates vendor lock-in, reduces the attack surface, and retains full freedom to optimize at the kernel level.

When a customer needs a custom attention mechanism or a novel quantization scheme, we can ship it in days, not quarters.

03 System Architecture

How the pieces fit together.

🖥️ **React 19 Frontend Dashboard** — Vite · TypeScript · Tailwind · Zustand · Framer Motion

↓ HTTP / WebSocket / SSE

🔗 **Node.js Gateway + BFF Proxy** — Express · Rate Limiting · Auth · WebSocket Hub

↓ REST / WebSocket / gRPC

🐍 **Python API Server** — FastAPI · OpenAI-compatible · Session Router · Tool Executor

↓ Model Forward Pass

🧠 **Aurelius Transformer (1.395B params)** — Pure PyTorch · GQA · RoPE · YaRN · SwiGLU · RMSNorm

The Transformer Core

SPECIFICATION	VALUE	BUSINESS IMPACT
Parameters	1.395 billion	Large enough for complex reasoning, small enough to run on commodity hardware
Architecture	Decoder-only transformer	Industry standard — proven at scale by GPT, Claude, and Llama families
Context Window	8,192 tokens (extendable to 128K+ via YaRN)	Process entire documents, multi-turn conversations, and code repositories
Vocabulary	128,000 tokens	Efficient compression of multilingual text, code, and technical notation
Attention	Grouped Query Attention (GQA)	2× faster inference vs. standard multi-head attention, lower memory footprint
Position Encoding	RoPE (θ = 500K) with YaRN	Stable long-context extrapolation without retraining
Activation	SwiGLU	State-of-the-art feed-forward efficiency used in PaLM and Llama-2
Normalization	RMSNorm (pre-norm)	Faster convergence, better stability at scale

Family Architecture

The `FamilyManifest` + `ModelVariant` pattern is Aurelius's answer to model proliferation. Instead of maintaining separate codebases for a base model, a chat model, a coding model, and a safety model, all variants share one core and specialize through configurable heads, tokenizers, and prompt templates. This reduces maintenance burden by an order of magnitude and lets customers hot-swap personas without reloading weights.

Base Variant

General-purpose autoregressive language modeling. The foundation for all other variants.

Chat Variant

Instruction-tuned with ChatML, Llama-3, and proprietary templates. Multi-turn conversation with tool awareness.

Coding Variant

Fill-in-the-middle (FIM) support, AST-aware tooling, and patch synthesis for software engineering workflows.

Long-Context Variant

YaRN-extended RoPE, attention sinks, and StreamingLLM for 128K+ token contexts.

Retrieval Variant

BM25 + dense hybrid search, citation tracking, and RAG-optimized prompt assembly.

Safety Variant

Constitutional AI filtering, refusal training, and red-team hardened output guardrails.

04 Frontend Dashboard

The face of the platform — designed for operators, not just developers.

The Aurelius dashboard is a single-page application built with React 19, TypeScript, Vite, and Tailwind CSS. It serves as a "Mission Control" for the entire AI platform, giving non-technical stakeholders visibility into system health, agent activity, and conversational memory.

15 Application Pages

PAGE	PURPOSE	KEY FEATURE
Dashboard	System-wide "Mission Control" overview	Live charts, agent status, activity feed, export
Chat	Interactive agent conversation	SSE streaming, slash commands, persistent history
Notifications	Hermes notification center	Bulk actions, push notifications, priority filtering
Skills	Plugin & skill management	Favorites, risk scoring, live execution
Workflows	Orchestration monitor	Progress bars, start/stop controls
Memory	Layered memory browser	Search, importance scores, associative recall
Agents	Agent comparison & detail	Side-by-side metrics, color theming
Models	Model load/unload control	Parameter display, state transitions
Training	Training run monitor	Sparkline charts, epoch tracking
Logs	System log viewer	Level filtering, pagination, auto-refresh
Health	Health check dashboard	Live latency tests, component status
API Docs	Interactive documentation	Copy-to-clipboard, collapsible categories
Settings	Platform configuration	Dirty-state tracking, validation, import/export
Tasks	Scheduled cron jobs	Resume/pause, manual trigger

User Experience Highlights

Keyboard-First Navigation

Cmd+K command palette, / for global search, ? for shortcuts, G+[key] for page jumps. Power users never need a mouse.

Progressive Web App

Installable on desktop and mobile. Service worker for offline resilience. System-aware dark/light theme toggle.

Real-Time Everywhere

SSE for chat streaming, polling for dashboards (5–15s intervals), WebSocket for bi-directional agent control.

Enterprise Security

Session auto-lock after inactivity, API key auth, secure context detection, license activation gate.

05 Backend & API Infrastructure

The engine room — where requests become intelligence.

Aurelius runs on a three-tier backend: a React frontend, a Node.js gateway/BFF layer, and a Python API server. This separation lets each tier scale independently and fail gracefully.

Python API Surface (30+ Endpoints)

🤖 Agent Control

List agents, send commands, change state (Active/Idle/Error), query per-agent activity feeds.

🔧 Skills & Workflows

Execute skills with variable injection, trigger workflows, monitor execution status in real time.

💬 Conversations

OpenAI-compatible `/v1/chat/completions`, WebSocket streaming, multi-turn history, tool calling.

📊 Observability

500-record log ring buffer, health checks, SSE event stream, notification preferences, system stats.

🧠 Memory

Layered memory browser (L0–L4), semantic search, entry creation, importance scoring, access analytics.

🔐 Security

Optional API-key auth, license validation, rate limiting (100 rps / burst 200), HMAC timing-safe comparison.

Node.js Gateway Layer

The gateway sits between the React app and the Python API, handling cross-cutting concerns so the Python layer can focus on AI logic.

Auth Middleware

JWT validation, API key proxying, session management. Fail-closed defaults.

Rate Limiting

Token-bucket algorithm with per-key, per-IP, per-route, and global chains.

WebSocket Hub

Channel-based broadcasting. Chat rooms, agent status feeds, system events.

Rust Native Modules

Three Rust crates compiled with LTO and single codegen unit provide performance-critical paths that Python cannot match.

data-engine

In-memory CRUD engine for agents, notifications, memory, and workflows.

token-counter

Unicode-aware token estimation and budget

usage-store

Persistent event logging via `redb` embedded KV store.

NAPI-RS binding for Node.js.

truncation. 10× faster than Python equivalents.

PyO3 binding for Python integration.

06 AI Capabilities

What the model can do, and how it does it safely.

Agent Framework

Aurelius is not just a chatbot — it is an agentic system that can plan, act, observe, and reflect. The ReAct (Reasoning + Acting) loop enables multi-step problem solving with tool use.

ReAct Loop

Plan → Act → Observe → Reflect. Per-tool wall-clock timeouts, argument validation via Python introspection, and budget-bounded termination prevent runaway execution.

Tool Calling

Parses both XML (Anthropic-style) and JSON (OpenAI-style) tool calls. Injection-hardened state-machine scanning prevents prompt injection via tool outputs.

Multi-Step Chaining

Tools can invoke other tools. Dependency DAGs with transitive skip logic. Parallelizable groups extracted automatically.

Error Recovery

Per-step exception handling with retry logic. Graceful degradation when tools fail or time out.

Memory System

Most LLMs are stateless — they forget everything between sessions. Aurelius maintains a layered semantic memory that persists across conversations and evolves over time.

LAYER	TYPE	LIFESPAN	USE CASE
L0 — Sensory	Raw input buffer	Single turn	Immediate context for the current query
L1 — Working	TTL + LRU cache	Minutes to hours	Active conversation threads, recently used facts
L2 — Episodic	Event sequences with importance scores	Days to weeks	Conversation history, decision trails, user preferences
L3 — Semantic	Concept graph + associative links	Months	Domain knowledge, entity relationships, learned patterns
L4 — Procedural	Skill recipes + workflow templates	Indefinite	How-to knowledge, standard operating procedures

Safety & Alignment

- ✓ **Jailbreak Detection** — 6-category regex ensemble with risk scoring
- ✓ **Red-Team Probes** — 6-category adversarial test suite with refusal detection

- ✓ **Prompt Injection Scanner** — Detects indirect injection via HTML, homoglyphs, zero-width characters
- ✓ **Output Filtering** — PII redaction, email/phone/SSN masking, API key detection
- ✓ **Constitutional AI** — Self-critique and revision based on principled rules
- ✓ **Hallucination Guard** — Claim verification against retrieval sources
- ✓ **Harm Taxonomy** — 7-category content classification with confidence scoring
- ✓ **Canary Tokens** — One-shot exfiltration detection in model outputs

07 Security & Trust

Enterprise-grade defense in depth.

Security is not a feature added at the end — it is woven into every layer of Aurelius. The project maintains a full CVE ledger, a STRIDE threat model, and continuous static analysis.

28	0	0	24
CVES TRACKED	OPEN CRITICAL	OPEN HIGH	SECURITY MODULES

Defense-in-Depth Controls

THREAT CATEGORY	CONTROL	STANDARD
Supply Chain	Hash-pinned lockfiles, pip-audit, SBOM generation	NIST SSDF
Deserialization	All `torch.load()` calls use `weights_only=True`	CWE-502
Path Traversal	Regex-safe ID validation + resolved-path jail	CWE-22
DoS	1 MiB body cap, token-bucket rate limiter (100 rps, burst 200)	CWE-400
Authentication	HMAC timing-safe comparison, SHA-256 hashed keys, fail-closed default	CWE-287
Injection	Bounded regex quantifiers, no DOTALL on untrusted input	CWE-94
Exfiltration	Canary token pipeline, one-shot detection	MITRE ATT&CK T1041
Archive Safety	Zip-slip, tar-slip, bomb, and symlink guards	CWE-22/400
Sandbox Escape	Deny-list blocks `getattr`/`setattr`/`type` traversal	CWE-250
Adversarial Robustness	GCG adversarial suffix search, gradient inversion defense	MLSec

MITRE ATT&CK Integration

The security subsystem includes a full MITRE ATT&CK taxonomy classifier covering all 12 enterprise tactics and 81 techniques. This enables security operations teams to map AI-generated threats to industry-standard frameworks and integrate Aurelius into existing SIEM workflows.

YARA-Like Rule Engine

Text, hex, and regex string matching with AND/OR/NOT logic, count operators, and

IOC Extractor

Automatically extracts IPs, domains, URLs, emails, hashes, CVEs, registry keys, and

filesize constraints for custom threat detection.

Bitcoin addresses from unstructured text with defang/refang support.

Continuous Security Monitoring

Every commit triggers Bandit static analysis, pip-audit dependency scanning, and Ruff security linting. The CI pipeline fails on any new High-severity finding. Red-team scripts run adversarial probes against the model before every release.

08 Deployment & DevOps

From laptop to data center.

Aurelius is designed to run anywhere: a developer's laptop, a private cloud, an air-gapped data center, or a Kubernetes cluster. The deployment artifacts are as first-class as the source code.

Deployment Options

🐳 Docker Compose

Three-tier stack (Python API + Node BFF + Nginx frontend) with optional Redis cache. Resource limits: 4 CPU / 16 GB RAM for the AI service.

⎈ Kubernetes

Full Helm chart with Deployment, Service, and HPA templates. Defaults: 1 replica, 8 GB request / 16 GB limit. GPU node affinity and autoscaling optional.

🦙 Ollama

GGUF conversion scripts for local desktop deployment. Run on Apple Silicon, consumer GPUs, or CPU with quantized weights.

☁️ Cloud-Native

Multi-environment Docker Compose with GPU overrides, nginx reverse proxy, and health-checked rolling updates.

CI/CD Pipeline

STAGE	TOOLS	GATE
Test	pytest (Python 3.12 & 3.13), Vitest (frontend)	All tests must pass
Lint	Ruff (Python), ESLint (TypeScript)	Zero lint errors
Security	Bandit, pip-audit	Zero High findings
Build	Vite (frontend), Docker Buildx	Successful artifact generation
Deploy	GHCR push, Helm upgrade	Tag-based or main-branch trigger

Observability Stack

- ✓ **Health Probes** — Readiness and liveness checks on every service
- ✓ **Metrics** — p50/p95/p99 latency histograms, throughput, error rates
- ✓ **Tracing** — Jaeger-format distributed trace export
- ✓ **Log Aggregation** — 500-record ring buffer with level filtering
- ✓ **SLO Tracking** — Burn-rate alerting for availability targets
- ✓ **Anomaly Detection** — z-score, IQR, and rolling-window outlier detection
- ✓ **Dashboard Export** — JSON, Prometheus, InfluxDB line protocol, CSV
- ✓ **Capacity Planning** — Linear regression trend analysis with days-to-limit forecasting

09 Intellectual Property Strategy

Open usage, protected architecture.

Aurelius uses a novel licensing model that balances ecosystem growth with competitive moat. The software is free to use, modify, and distribute. The architecture is not.

The Aurelius Open License

Grants broad rights to use, copy, modify, merge, publish, distribute, sublicense, and sell the Software. Requires attribution. The underlying Aurelius architecture — including the agent orchestration model, memory hierarchy, and core design patterns — remains the exclusive intellectual property of Aurelius Systems, Inc. Users may not claim ownership of the architecture or file patents on it. The Aurelius name and logo are reserved trademarks.

What This Means for the Business

✅ Community Adoption

Developers can use Aurelius for free, which drives word-of-mouth, bug reports, and ecosystem plugins. Open-source is our best marketing channel.

✅ Enterprise Licensing

Large customers who need indemnification, support SLAs, or custom variants pay for commercial licenses. The open core drives inbound enterprise leads.

🔒 Architecture Protection

Competitors cannot simply fork the codebase and claim the architecture as their own. The IP clause preserves our technical differentiation.

🔒 Trademark Control

Only official distributions can use the Aurelius brand. This prevents dilution and ensures customers know what they are getting.

Confidential Materials

The following categories are classified as **internal use only** and are not part of the open-source distribution:

- ✓ Training datasets and proprietary corpora
- ✓ Pretraining and fine-tuning configurations
- ✓ Model checkpoints and trained weights
- ✓ Experiment logs and harvest journals
- ✓ Data extraction and processing pipelines
- ✓ Hyperparameter schedules and recipes
- ✓ Internal benchmark results and evaluations
- ✓ Strategic roadmaps and planning documents

10 Competitive Position

Where Aurelius fits in the landscape.

Capability Comparison

CAPABILITY	AURELIUS	OPENAI API	ANTHROPIC CLAUDE	LOCAL LLMS (OLLAMA)
Full source ownership	✅ Yes	❌ No	❌ No	⚠️ Partial
On-premise / air-gapped	✅ Native	❌ No	❌ No	✅ Yes
Custom architecture research	✅ Native	❌ No	❌ No	⚠️ Limited
Agentic tool use + memory	✅ Built-in	⚠️ Add-ons	⚠️ Add-ons	⚠️ Community
Enterprise security hardening	✅ 28 CVEs closed	✅ SOC 2	✅ SOC 2	❌ Minimal
Modern web dashboard	✅ React 19 SPA	❌ No	❌ No	❌ CLI only
Multi-variant model family	✅ FamilyManifest	❌ Separate models	❌ Separate models	❌ Manual
Cost per 1M tokens	✅ Hardware only	❌ $2–$15	❌ $3–$75	✅ Hardware only

Key Differentiators

1. Vertical Integration

From transformer weights to pixel-perfect UI, every layer is controlled. No black boxes, no vendor lock-in, no surprise API deprecations.

2. Security by Design

28 tracked CVEs, zero open High findings, full STRIDE threat model, MITRE ATT&CK integration, and adversarial robustness testing. Enterprise security teams can trust this.

3. Research Agility

150+ architecture modules (SSMs, MoE, diffusion heads) let researchers prototype novel ideas without migrating between frameworks. Ideas become products faster.

4. Total Cost of Ownership

No per-token pricing. Run on existing hardware — Apple Silicon, consumer GPUs, or cloud instances. The only cost is infrastructure, not API credits.

> *"Aurelius is what you get when you combine the research depth of an academic lab with the product discipline of an enterprise SaaS company — and then make it runnable on a laptop."*

11 Development Velocity

The numbers behind the platform.

157+	1,720+	32,400+	33
IMPLEMENTATION CYCLES	PYTHON MODULES	PASSING TESTS	SUB-SYSTEMS

Code Distribution

Quality Metrics

Test Coverage

Every module ships with 10–16 tests covering shape validation, gradient flow, determinism, edge cases, and numerical stability.

Static Analysis

Ruff linting with security rules (E, F, I, UP, S). Bandit continuous scanning. pip-audit dependency vulnerability checks.

Registry Isolation

pytest autouse fixtures snapshot and restore module-level singleton registries before every test, eliminating cross-test contamination.

Technology Footprint

LAYER	PRIMARY LANGUAGES	KEY FRAMEWORKS
AI Core	Python 3.12+	PyTorch 2.4+ (pure, no wrappers)
API Server	Python, TypeScript	FastAPI, Express, custom HTTP
Gateway / BFF	TypeScript	Express, WebSocket, Redis
Frontend	TypeScript	React 19, Vite, Tailwind, Zustand
Native Performance	Rust	NAPI-RS, PyO3, redb
Data & Migrations	Python, SQL	SQLAlchemy, Alembic

LAYER	PRIMARY LANGUAGES	KEY FRAMEWORKS
Infrastructure	YAML, Shell	Docker, Kubernetes, Helm, nginx

12 Investment Thesis

Why Aurelius represents a compelling opportunity.

Market Context

The generative AI market is projected to exceed $1.3 trillion by 2032. Yet most enterprises face a painful choice: rent intelligence from OpenAI/Anthropic and surrender data sovereignty, or cobble together open-source tools and accept operational fragility. Aurelius offers a third path: full ownership with production polish.

🌊 The Sovereignty Wave

Regulations (GDPR, HIPAA, SEC cybersecurity rules) and geopolitical tensions are driving demand for on-premise AI. Aurelius is built for air-gapped deployment from day one.

💰 The Cost Revolution

At $2–$15 per million tokens, API costs scale linearly with usage. A busy enterprise can spend six figures monthly. Aurelius shifts that to a fixed hardware cost with no marginal token pricing.

🔬 The Research Premium

Organizations that can customize their models — swap attention mechanisms, add safety layers, or compress for edge devices — gain compounding advantages. Framework-dependent competitors cannot match this agility.

🛡️ The Security Imperative

AI systems are attack surfaces. Prompt injection, model extraction, and data poisoning are real threats. Aurelius's 24 security modules and adversarial testing suite make it the safest choice for regulated industries.

Revenue Model

STREAM	DESCRIPTION	TARGET CUSTOMERS
Enterprise License	Commercial use with indemnification, support SLA, and custom variants	Fortune 500, regulated industries
Professional Services	Custom model training, architecture consulting, integration support	Mid-market, government
Managed Hosting	Fully managed Aurelius deployment in customer's cloud account	Teams without ML ops expertise
Training & Certification	Workshops on safe AI deployment, red-team exercises, architecture deep-dives	Security teams, auditors

The Open-Core Flywheel

By open-sourcing the platform under the Aurelius Open License, we turn every developer into a potential advocate, contributor, and lead. The open core drives awareness and adoption. The

proprietary architecture IP and commercial licenses capture value. This is the same model that built MongoDB, Elastic, and Confluent into multi-billion-dollar companies.

13 Contact & Next Steps

How to engage with Aurelius Systems.

📧 General Inquiries

hello@aurelius.systems

For partnership discussions, press inquiries, and general questions.

🔐 Security & Compliance

security@aurelius.systems

For vulnerability reports, audit requests, and security assessments.

⚖️ Legal

legal@aurelius.systems

For licensing negotiations, IP matters, and contractual questions.

💼 Investment

investors@aurelius.systems

For funding discussions, due diligence, and board materials.

Suggested Next Steps

AUDIENCE	RECOMMENDED ACTION	TIMELINE
CTO / Engineering Lead	Clone the repo, run `aurelius serve`, and evaluate the API compatibility with your existing stack	1 day
CISO / Security Team	Review the threat model, CVE ledger, and run Bandit + pip-audit against the codebase	2–3 days
CEO / Board	Schedule a live demo of the dashboard, agent tool use, and memory system	1 hour
Investor	Request access to the data room for financial projections, cap table, and technical deep-dive	Upon NDA
Enterprise Buyer	Pilot program: deploy on-premise with your data, measure against incumbent API costs	2–4 weeks

About This Document

This executive overview was generated on April 26, 2025, based on an extensive review of the Aurelius codebase. Training datasets, proprietary configurations, and internal experiment logs have been intentionally excluded in accordance with the Aurelius confidentiality policy. For access to confidential materials, contact legal@aurelius.systems.

github.com/S3nna13/Aurelius · hello@aurelius.systems

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